

UNIT-7

Number Theoretic Algorithms:

Small easier unit so read everything instead of imp only ✓

Number Theoretic Notations:

i) Divisibility and Divisors: The notation $d|a$ (read "d divides a") means that $a=kd$ for some integer k . If $d|a$, then we say that 'a' is a multiple of d . Every integer divides 0. If $a>0$ and $d|a$, then $|d| \leq |a|$. If $d|a$ and $d \geq 0$, we say that d is a divisor of a . A divisor of an integer 'a' is at least 1 but not greater than $|a|$.

For example: The divisors of 24 are 1, 2, 3, 4, 6, 8, 12 and 24.

trivial divisors of 24: 1 and 24

non-trivial divisors of 24: 2, 3, 4, 6, 8 and 12

1 and the no. itself

other remaining ones than that of trivial

Prime and Composite Numbers:

An integer $a>1$ whose only divisors are the trivial divisors (i.e. 1 and 'a' itself only) is said to be a prime number or simply prime.

An integer $a>1$ that is not prime is said to be a composite number. The integer 1 is said to be a unit and is neither prime nor composite. Similarly, the integer 0 and all negative integers are neither prime nor composite.

Common Divisors and Greatest Common Divisors:

If 'd' is a divisor of 'a' and also divisor of 'b', then 'd' is a common divisor of 'a' and 'b'. Note that 1 is a common divisor of any two integers. An important property of common divisors is that:

→ if $d|a$ and $d|b$ then, $d|(a+b)$ and $d|(a-b)$.

→ if $d|a$ and $d|b$ then, $d|(ax+by)$

The greatest common divisor $\gcd(a, b)$ is the largest of the common divisors of a and b . We define $\gcd(0, 0)$ to be 0.

→ For any integers a and b if $d|a$ and $d|b$ then, $d|\gcd(a, b)$.

→ For all integers a and b and any non-negative integer n ,

$$\gcd(an, bn) = n \gcd(a, b)$$

→ For all positive integers n, a , and b , if $n|ab$ and $\gcd(a, n)=1$, then $n|b$.

Relatively Prime Integers:

Two integers a, b are said to be relatively prime if their only common divisor is 1, that is if $\gcd(a, b) = 1$.

→ For any integers a, b , and p , if both $\gcd(a, p) = 1$ and $\gcd(b, p) = 1$ then, $\gcd(ab, p) = 1$.

→ For all primes p and all integers a, b if $p \nmid ab$, then $p \nmid a$ or $p \nmid b$ or both.

Euclid's Algorithm for Solving Modular Linear Equations: [Imp]

Euclidean algorithm is an efficient method for computing the greatest common divisor of two numbers.

Algorithm:

EUCLID(a, b)

{ if $b = 0$
 return a ;

else

 return EUCLID($b, a \bmod b$)

}

Analysis: Since it is recursive algorithm so we need to find their recurrence relation. Since every time the problem is divided into two parts one is b and another is $a \bmod b$.

Thus the size of sub-problem = $\frac{n}{2}$

Dividing and merging time = constant = $O(1)$

Thus, recurrence relation is, $T(n) = T(n/2) + O(1)$

By solving this we get, $T(n) = O(\log n)$.

Example: Find $\gcd(30, 21)$ by Euclid's Algorithm.

Solution

EUCLID(30, 21)
= EUCLID(21, $30 \bmod 21$)

= EUCLID(21, 9)

= EUCLID(9, $21 \bmod 9$)

= EUCLID(9, 3)

= EUCLID(3, $9 \bmod 3$)

= EUCLID(3, 0)

Since $b = 0$, so return a

= 3

∴ $\gcd(30, 21) = 3$.

* Extended Euclid's Algorithm for solving Modular Linear Equation: [Imp]

Extended Euclid's Algorithm is an extension to Euclid's algorithm which computes the coefficients of Bezout's identity, (which are integers x and y) in addition to the greatest common divisors of integers a and b such that:

$$d = \gcd(a, b) = ax + by$$

where, x and y may be zero or negative.

Algorithm:

EXTENDED-EUCLID(a, b)

{ if $b = 0$

return ($a, 1, 0$);

(d', x', y') $\leftarrow$ EXTENDED-EUCLID($b, a \bmod b$)

(d, x, y) $\leftarrow$ ($d', y', x' - \text{floor}(a/b)y'$)

return (d, x, y);

}

Analysis: Same as Euclid's algorithm i.e., $T(n) = O(\log n)$.

Example: Find GCD(161, 28) and value of x and y by using extended Euclidean's algorithm.

Solution:

We have, $a = 161$, $b = 28$ and $\gcd(a, b) = ax + by$.

Let's define following three equations;

$$r = r_1 - q * r_2$$

$$x = x_1 - q * x_2$$

$$y = y_1 - q * y_2$$

Consider $a = r_1$ and $b = r_2$.

formulas that will be used to calculate r, x, y in below table

always initialize
 $x_1 = 1, x_2 = 0$
& $y_1 = 0, y_2 = 1$

remainder

q	r_1	r_2	r	x_1	x_2	x	y_1	y_2	y
5	161	28	21	1	0	1	0	1	-5
1	28	21	7	0	1	-1	1	-5	6
3	21	7	0	1	-1	4	-5	6	-23
	7	0		-1	4		6	-23	

$$x = -1, y = 6$$

Since: $ax + by = \gcd(a, b)$

$$\text{or, } \{161 * (-1)\} + (28 * 6) = 7$$

$$\text{or, } 7 = 7$$

stop at step 4
so last step shift only

⊗ Miller-Rabin Randomized Primality Test:

This test algorithm test determines whether a given number is prime or not.

Algorithm:

/* It returns false if n is composite and returns true if n is probably prime. k is an input parameter that determines accuracy level. Higher value of k indicates more accuracy. */

bool IsPrime(int n , int k)

1) Handle base cases for $n < 3$

2) If n is even, return false.

3) Find an odd number d such that $n-1$ can be written as $d \cdot 2^r$.
Note that since n is odd, $(n-1)$ must be even and r must be greater than 0.

4) Do following k times

if (millerTest(n, d) == false)
return false.

5) Return true.

bool millerTest(int n , int d)

1) Pick a random number a in range $[2, n-2]$

2) Compute: $x = \text{pow}(a, d) \% n$

3) If $x == 1$ or $x == n-1$, return true.

// Below loop mainly runs ' $r-1$ ' times.

4) Do following while d doesn't become $n-1$.

a) $x = (x * x) \% n$.

b) If $(x == 1)$ return false.

c) If $(x == n-1)$ return true.

Example: Input: $n=13, k=2$.

1). Compute d and r such that $d \cdot 2^r = n-1$,
 $d=3, r=2$.

2) Call millerTest k times.

1st Iteration:

1) Pick a random number ' a ' in range $[2, n-2]$

Suppose $a=4$

2). Compute: $x = \text{pow}(a, d) \% n$

$$x = 4^3 \% 13 = 12$$

3). Since $x = (n-1)$, return true.

2nd Iteration:

1) Pick a random number ' a ' in range $[2, n-2]$

Suppose $a=5$

2) Compute: $x = \text{pow}(a, d) \% n$

$$x = 5^3 \% 13 = 8$$

3) x neither 1 nor 12.

4) Do following $(r-1)=1$ times

$$a) x = (x * x) \% 13 = (8 * 8) \% 13 = 12$$

b) Since $x = (n-1)$, return true.

Since both iterations return true, we return true.

⊗ Chinese Remainder Theorem:

The Chinese remainder theorem states that if one knows the remainders of the Euclidean division of an integer n by several integers, then one can determine uniquely the remainder of the division of n by the product of these integers, under the condition that the divisors are pairwise co-prime.

Statement: If $m_1, m_2, \dots, m_k$ are pairwise relatively prime positive integers, and if $a_1, a_2, \dots, a_k$ are any integers, then the simultaneous congruences

$$x \equiv a_1 \pmod{m_1},$$

$$x \equiv a_2 \pmod{m_2}, \dots$$

$$x \equiv a_k \pmod{m_k} \text{ have a solution, and the solution is unique modulo } m.$$

Here we need to calculate x with the help of following formulas:

$$x = (M_1 x_1 a_1 + M_2 x_2 a_2 + \dots + M_k x_k a_k) \pmod{M}$$

$$M = m_1 \cdot m_2 \cdot \dots \cdot m_k$$

$$M_j = \frac{M}{m_j}$$

$$\& M_j x_j = 1 \pmod{m_j}$$

Example: Solve following congruences by using Chinese remainder theorem.

$$x \equiv 1 \pmod{5}$$

$$x \equiv 1 \pmod{7}$$

$$x \equiv 3 \pmod{11}$$

Solution:

we have, $a_1 = 1, a_2 = 1, a_3 = 3$

$$m_1 = 5, m_2 = 7, m_3 = 11$$

$$x = (M_1 x_1 a_1 + M_2 x_2 a_2 + M_3 x_3 a_3) \pmod{M}$$

$$M = m_1 \cdot m_2 \cdot m_3 = 5 \times 7 \times 11 = 385$$

$$M_1 = \frac{M}{m_1} = \frac{385}{5} = 77$$

$$M_2 = \frac{M}{m_2} = \frac{385}{7} = 55$$

$$M_3 = \frac{M}{m_3} = \frac{385}{11} = 35$$

Now, $M_1 x_1 = 1 \pmod{5}$

or, $77x_1 = 1 \pmod{5}$

or, $2x_1 = 1 \pmod{5}$

or, $x_1 = 3$

Since we have to put value of x_1 in such a way that (product of 2 and x_1) $\pmod{5}$ becomes equal to 1.
for eg. let we put 1 then, $(2 \times 1) \pmod{5} = 2$
Now we put 3 then, $2 \times 3 = 6 \pmod{5} = 1$
So, $x_1 = 3$

$M_2 x_2 = 1 \pmod{7}$

or, $55x_2 = 1 \pmod{7}$

or, $6x_2 = 1 \pmod{7}$

or, $x_2 = 6$

$M_3 x_3 = 1 \pmod{11}$

or, $35x_3 = 1 \pmod{11}$

or, $2x_3 = 1 \pmod{11}$

or, $x_3 = 6$

random multiply

$6 \times 1 = 6$

$6 \times 2 = 12$

$6 \times 3 = 18$

$6 \times 4 = 24$

$6 \times 5 = 30$

$6 \times 6 = 36$

mod 7 value

6 ($\neq 1$)

5 ($\neq 1$)

4 ($\neq 1$)

3 ($\neq 1$)

2 ($\neq 1$)

1 ($= 1$)

Hence, $x_2 = 6$ or
putting 6 and doing
mod we got 1

$$\therefore x = (M_1 x_1 a_1 + M_2 x_2 a_2 + M_3 x_3 a_3) \pmod{M}$$

$$= (77 \times 3 \times 1 + 55 \times 6 \times 1 + 35 \times 6 \times 3) \pmod{385}$$

$$= (231 + 330 + 630) \pmod{385}$$

$$= 1191 \pmod{385}$$

$$= 36 \text{ Ans.}$$

Also, we can test the solution as;

$$36 \pmod{5} = 1$$

$$36 \pmod{7} = 1$$

$$36 \pmod{11} = 3$$